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Unit Name: Machine Learning

Unit Code: SIT720

Assignment: 8.1 DISTINCTION TASK (Machine Learning Mini Project)

1. Problem Definition and Data Collection

The objective of this project is to design an algorithmic method that will help in forecasting the sale price of Sydney properties using machine learning. Sale Price was chosen as the prediction variable, while property attributes like suburb, bedrooms, bathrooms, parking, land size and property type were chosen as the predictor variables.

Three Sydney suburbs, namely Blacktown, Liverpool, and Parramatta, have been chosen to give properties from varied local housing market conditions. In total, 121 sold residential properties were obtained from the Sydney suburbs, with at least 30 properties from each of the chosen suburbs. The dataset contains nine variables, which are Suburb, Sale Price, Bedrooms, Bathrooms, Parking, Land Size, Property Type, Sale Date and Property Address.

Some missing data were found in the dataset. Missing values were found in Bedrooms (2), Bathrooms (3), Parking (8), and Land Size (39). The highest number of missing values was found in Land Size. Missing values were not found in Sale Price, Suburb, Property Type, Sale Date and Property Address.

Some issues should be taken into account in the analysis of the obtained results. First of all, the data set is rather small and consists of only three suburbs, so the results may not fully characterize the Sydney real estate market. Moreover, the set of variables obtained does not include all factors that affect property price, such as the condition of the house, its interior design, exact location, distance from transport and educational institutions, views, etc.

It means that the data collection process has resulted in the creation of a good data set for illustration of the machine learning algorithm, but some limitations should be considered during interpretation of the results obtained.

	Suburb	Sale Price	Bedrooms	Bathrooms	Parking	Land Size	Property Type	Sale Date	Property Address
0	Blacktown	\$980,000	3.0	1.0	2.0	613 m ²	House	2026-09-18	49 Pendant Avenue
1	Blacktown	\$1,350,000	5.0	3.0	3.0	338 m ²	House	2026-09-17	40 Stephen Street
2	Blacktown	\$900,000	3.0	1.0	3.0	556 m ²	House	2026-09-17	199 Flushcombe Road
3	Blacktown	\$1,105,000	3.0	1.0	3.0	739.8 m ²	House	2026-09-16	6 Oregon Street
4	Blacktown	\$1,135,000	3.0	2.0	4.0	765 m ²	House	2026-09-15	7 Highlands Crescent

2. Data Understanding and Feature Engineering

There are 121 property records in the dataset for the suburbs of Blacktown, Liverpool and Parramatta. The variable Sale Price is the target variable in the dataset while other variables include Bedrooms, Bathrooms, Parking, Land Size, Property Type, Sale Date and Property Address. Missing values were present in some of the variables like Land Size, Parking, Bedrooms and Bathrooms.

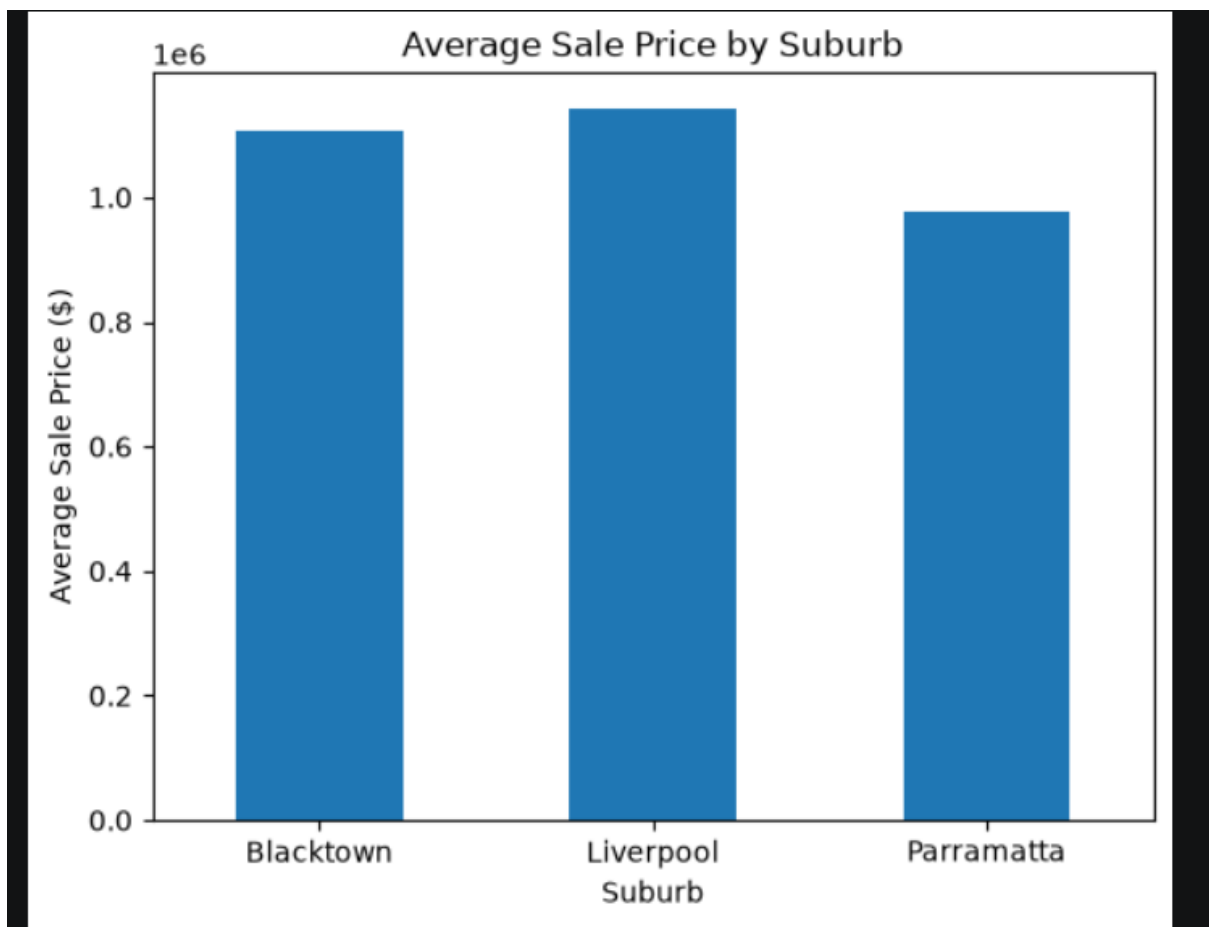
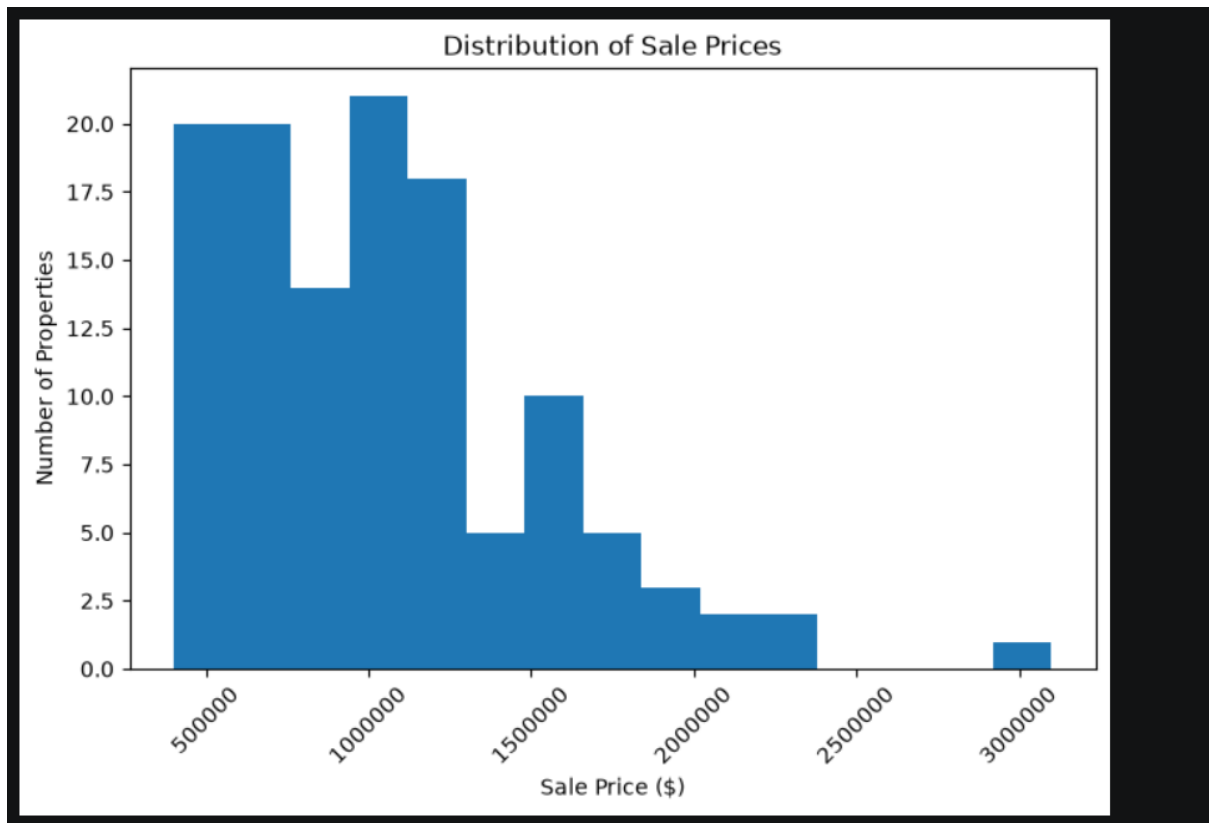
Sale prices of properties were different for the three suburbs. Blacktown had a sale price of about \$1,107,600 while Liverpool had a sale price of about \$1,142,233 and the lowest sale price was of Parramatta which was about \$976,295.

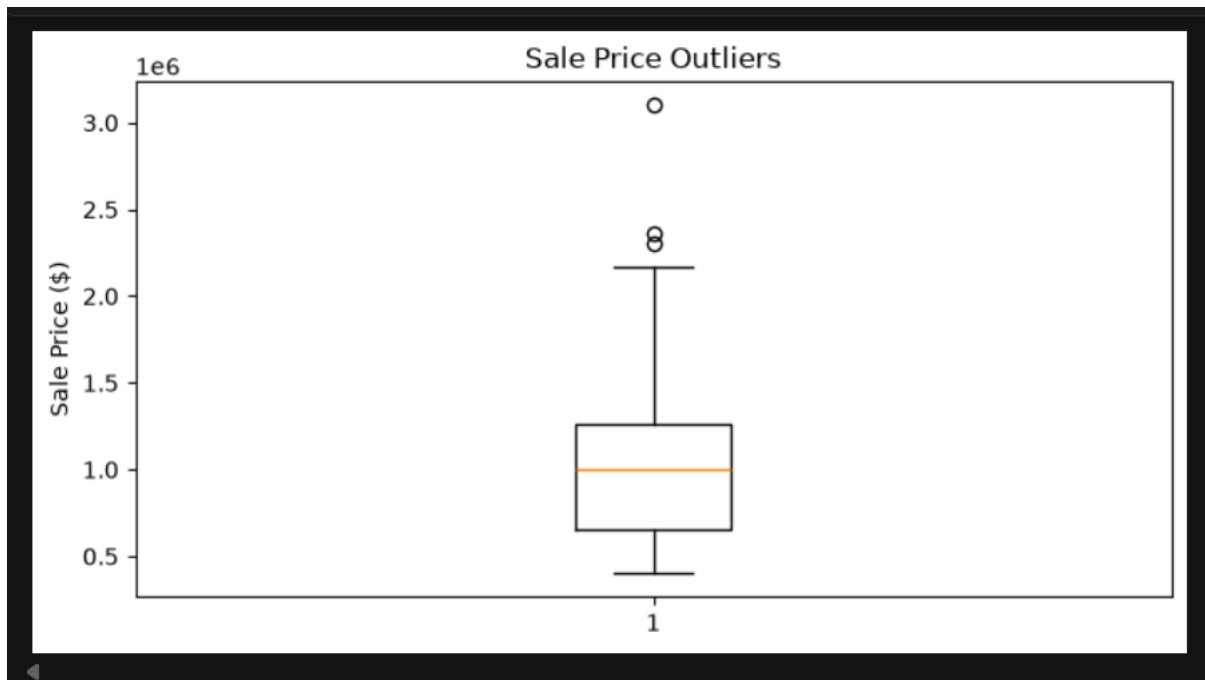
The notebook also included a box plot of Sale Price in order to visualize its distribution and detect unusual values. There is a variation in property prices and the presence of high-valued observations in the Sale Price data set. The notebook did not compute the exact number of statistical outliers, thus, no number of outliers is specified.

Prior to feature engineering, the relationships between numerical variables and Sale Price were evaluated based on correlation values. The highest correlation of Sale Price was found for Bedrooms (0.704), then Land Size (0.681) and Bathrooms (0.516). Therefore, these three features are the strongest variables that influence the price of properties. The correlation between Bedrooms and Bathrooms was also quite high (0.880).

Therefore, it can be stated that the decision to engineer a new feature based on the above analysis was correct. A feature Total Rooms was engineered by adding Bedrooms and Bathrooms since they were positively correlated with Sale Price.

This analysis was restricted by lack of property information like “Land Size” and “Parking.” Due to the relatively small size of data set, the missing information was dealt with before running the analysis since deleting many observations was not necessary. In summary, the exploratory analysis formed the base for selection of variables for the regression analysis.





```
Suburb
Blacktown      1.107600e+06
Liverpool      1.142233e+06
Parramatta     9.762951e+05
Name: Sale Price, dtype: float64
```

3. Model Development and Evaluation

Three regression models were built: Linear Regression, Random Forest Regression, and Gradient Boosting Regression. The use of Linear Regression was chosen as a baseline model while Random Forest and Gradient Boosting Regression were used to account for more complicated dependencies between predictors and target variable.

Before evaluating the models, Random Forest was expected to perform well because it can capture more complex relationships between the property features and sale price.

The evaluation of all three models was done by means of 5-fold cross-validation in terms of MAE, RMSE, and R^2 metrics. Linear Regression gave MAE = \$142,210, RMSE = \$190,626, and $R^2 = 0.777$. Random Forest gave MAE = \$139,988, RMSE = \$188,285, and $R^2 = 0.808$. Gradient Boosting gave MAE = \$145,916, RMSE = \$200,338, and $R^2 = 0.781$.

According to these cross validation results, the best performing model is the Random Forest, and this model is chosen as the final model. When this model is evaluated using the test set, an MAE of \$199,371, RMSE of \$361,218 and R^2 of 0.640 is obtained.

This poor test-set performance compared to the cross validation results shows that this model does not generalise well on unseen properties. Also, the larger RMSE suggests that there were some properties that had larger prediction errors. This illustrates how challenging it is to predict housing prices when the data set used is small with limited property information.

The difference between the cross-validation and test-set performance suggests that the Random Forest model may have difficulty generalising to some unseen properties. However, the available results do not provide enough evidence to conclusively classify the model as underfitting or overfitting. The use of a more complex model can capture nonlinear relationships, but its performance still depends on the size and quality of the training data

In general, the Random Forest model showed the best cross validation performance amongst the three models and hence it was used in the later prediction phase.

```
{ 'Linear Regression': { 'MAE': np.float64(142210.00080368415),  
  'RMSE': np.float64(190626.17539859755),  
  'R2': np.float64(0.7766499001136284)},  
  'Random Forest': { 'MAE': np.float64(139987.96921593757),  
  'RMSE': np.float64(188284.92832413365),  
  'R2': np.float64(0.8076273437670738)},  
  'Gradient Boosting': { 'MAE': np.float64(145916.46432514692),  
  'RMSE': np.float64(200337.57011751193),  
  'R2': np.float64(0.7808353390481131)}}}
```

```
MAE: 199371.2525063825  
RMSE: 361217.67671140865  
R2: 0.6399868858491053
```

4. Investigating Prediction Failures

The five biggest prediction errors were examined in order to determine how the Random Forest model failed. The biggest prediction error was for 1 Leitz Street,

Liverpool, with an actual sale price of \$3,100,000 but a predicted price of around \$1,773,968, which resulted in an error of roughly \$1,326,032. This house was very different from many others because it had 24 bedrooms, 14 bathrooms, 6 car parking spaces and a land size of 1,358 m².

The second-biggest prediction error was for 3 Secant Street, Liverpool, with an actual price of \$2,000,000 and a predicted price of roughly \$1,220,028. The absolute error for this case was about \$779,973. This house had 4 bedrooms, 2 bathrooms, 2 car parking spaces and a land area of 676 m².

The third largest error was 84 Crimea Street, Parramatta which sold at \$2,170,000 against the predicted price of approximately \$1,462,945, an error of about \$707,055. It had 5 bedrooms, 3 bathrooms, 2 car parks and land size of 303 m².

The fourth largest error was 33 Rosehill St, Parramatta that sold at \$2,080,000 while the predicted price was approximately \$1,785,355, an error of about \$294,645. It had 4 bedrooms, 1 bathroom, 1 car park and land size of 691 m².

The fifth largest error was 610/36-46 Cowper Street, Parramatta with the actual price at \$575,000 and the model predicted approximately \$869,096, an error of approximately \$294,096. There were several missing features of the property such as number of bedrooms, bathrooms, parking, and land size.

From these mistakes, it is clear that this model might face difficulties while working with properties which are of unusual size or unusual in any way, and also while working with properties having incomplete data. The current set of features does not take into account all possible variables that could affect the sales price of a property, such as property condition, renovations, special characteristics of location or any other relevant context.

	Suburb	Sale Price	Bedrooms	Bathrooms	Parking	Land Size	Property Type	Property Address
47	Liverpool	3100000.0	24.0	14.0	6.0	1358.0	House	1 Leitz Street
31	Liverpool	2000000.0	4.0	2.0	2.0	676.0	House	3 Secant Street
118	Parramatta	2170000.0	5.0	3.0	2.0	303.0	House	84 Crimea Street
89	Parramatta	2080000.0	4.0	1.0	1.0	691.0	House	33 Rosehill St
64	Parramatta	575000.0	NaN	NaN	NaN	NaN	Apartment	610/36-46 Cowper Street

	Suburb	Bedrooms	Bathrooms	Parking	Land Size	Property Type	Property Address	Sale Price	ML Prediction
44	Liverpool	3.0	2.0	1.0	NaN	House	5/10 Christie Street	870000.0	1.091616e+06
47	Liverpool	24.0	14.0	6.0	1358.0	House	1 Leitz Street	3100000.0	1.773968e+06
4	Blacktown	3.0	2.0	4.0	765.0	House	7 Highlands Crescent	1135000.0	1.246938e+06
55	Liverpool	3.0	1.0	2.0	559.0	House	32 Gill Avenue	1160000.0	1.061636e+06
26	Blacktown	3.0	2.0	3.0	600.7	House	59 Lancaster Street	900000.0	1.015788e+06
64	Parramatta	NaN	NaN	NaN	NaN	Apartment	610/36-46 Cowper Street	575000.0	8.690963e+05
73	Parramatta	2.0	1.0	1.0	104.0	Unit	3/23 Good Street	569000.0	5.687982e+05
10	Blacktown	3.0	1.0	1.0	575.4	House	19 Stephen Street	1062500.0	9.417050e+05
40	Liverpool	3.0	2.0	1.0	233.0	Townhouse	4/44-46 Passefield Street	875000.0	1.007405e+06
108	Parramatta	3.0	2.0	2.0	204.0	Townhouse	10/43 Pemberton Street	1101500.0	1.113995e+06

	Suburb	Bedrooms	Bathrooms	Parking	Land Size	Property Type	Property Address	Sale Price	ML Prediction
44	Liverpool	3.0	2.0	1.0	NaN	House	5/10 Christie Street	870000.0	1.091616e+06
47	Liverpool	24.0	14.0	6.0	1358.0	House	1 Leitz Street	3100000.0	1.773968e+06
4	Blacktown	3.0	2.0	4.0	765.0	House	7 Highlands Crescent	1135000.0	1.246938e+06
55	Liverpool	3.0	1.0	2.0	559.0	House	32 Gill Avenue	1160000.0	1.061636e+06
26	Blacktown	3.0	2.0	3.0	600.7	House	59 Lancaster Street	900000.0	1.015788e+06
64	Parramatta	NaN	NaN	NaN	NaN	Apartment	610/36-46 Cowper Street	575000.0	8.690963e+05
73	Parramatta	2.0	1.0	1.0	104.0	Unit	3/23 Good Street	569000.0	5.687982e+05
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108	Parramatta	3.0	2.0	2.0	204.0	Townhouse	10/43 Pemberton Street	1101500.0	1.113995e+06

5. Human Judgement, Machine Learning and Large Language Models

Estimates were made using three different methods for ten properties in the held-out test set: a Random Forest method, a ChatGPT estimate, and a human estimate. The same data about each property was provided for the comparison.

The Random Forest predictions led to an average absolute error (MAE) of \$243,373.06 for the ten properties. The ChatGPT estimates led to an average absolute error (MAE) of \$102,000, while the human estimates led to an average absolute error (MAE) of \$137,000.

The conclusion is that there were significant differences between the three estimations for some properties. The advantage of the machine learning model is in training directly with the dataset that was collected; however, it can be impacted by the unique properties and lack of data. The ChatGPT estimates were made based on the provided property information, but they did not use the trained machine learning model using the project dataset. This is why those estimates have to be considered as experimental ones.

The above discussion demonstrates that automated prediction can offer a reproducible methodology, whereas human and LLM estimates include reasoning that is not expressed in numeric terms. It is important to note, however, that the

comparison includes only ten objects and should not be generalized outside this set.

	Suburb	Bedrooms	Bathrooms	Parking	Land Size	Property Type	Property Address	Sale Price	ML Prediction	ChatGPT Prediction	Human Prediction
44	Liverpool	3.0	2.0	1.0	NaN	House	5/10 Christie Street	870000.0	1.091616e+06	950000	900000
47	Liverpool	24.0	14.0	6.0	1358.0	House	1 Leitz Street	3100000.0	1.773968e+06	2500000	2000000
4	Blacktown	3.0	2.0	4.0	765.0	House	7 Highlands Crescent	1135000.0	1.246938e+06	1150000	1150000
55	Liverpool	3.0	1.0	2.0	559.0	House	32 Gill Avenue	1160000.0	1.061636e+06	1050000	1100000
26	Blacktown	3.0	2.0	3.0	600.7	House	59 Lancaster Street	900000.0	1.015788e+06	950000	950000
64	Parramatta	NaN	NaN	NaN	NaN	Apartment	610/36-46 Cowper Street	575000.0	8.690963e+05	600000	600000
73	Parramatta	2.0	1.0	1.0	104.0	Unit	3/23 Good Street	569000.0	5.687982e+05	570000	570000
10	Blacktown	3.0	1.0	1.0	575.4	House	19 Stephen Street	1062500.0	9.417050e+05	1000000	1000000
40	Liverpool	3.0	2.0	1.0	233.0	Townhouse	4/44-46 Passefield Street	875000.0	1.007405e+06	850000	850000
108	Parramatta	3.0	2.0	2.0	204.0	Townhouse	10/43 Pemberton Street	1101500.0	1.113995e+06	1050000	1100000

	Property Address	Sale Price	ML Error	ChatGPT Error	Human Error
44	5/10 Christie Street	870000.0	2.216160e+05	80000.0	30000.0
47	1 Leitz Street	3100000.0	1.326032e+06	600000.0	1100000.0
4	7 Highlands Crescent	1135000.0	1.119375e+05	15000.0	15000.0
55	32 Gill Avenue	1160000.0	9.836400e+04	110000.0	60000.0
26	59 Lancaster Street	900000.0	1.157875e+05	50000.0	50000.0
64	610/36-46 Cowper Street	575000.0	2.940963e+05	25000.0	25000.0
73	3/23 Good Street	569000.0	2.017857e+02	1000.0	1000.0
10	19 Stephen Street	1062500.0	1.207950e+05	62500.0	62500.0
40	4/44-46 Passefield Street	875000.0	1.324050e+05	25000.0	25000.0
108	10/43 Pemberton Street	1101500.0	1.249500e+04	51500.0	1500.0

```
ML Error      243373.061931
ChatGPT Error 102000.000000
Human Error   137000.000000
dtype: float64
```

Part 6 Final Deployment and Reflection

A simple web application for predicting the price of houses was created using Streamlit. The app loads the trained `housing_price_model.pkl` file and enables the user to enter house details like suburb, number of bedrooms, number of bathrooms, parking, landsize, and property type. Total number of rooms is calculated from bedrooms and bathrooms, and all the data is fed into the trained model. The prediction is outputted as the estimated sale price.

The application is easily executable using Streamlit with the given Python application file and the trained model file in the same directory. The user enters the details of the required property and then clicks on "Predict Sale Price".

In this project, we went through all steps of the machine learning process starting from the collection and cleaning of the housing data to feature engineering, model development, testing, and deployment. One of the problems encountered in this project included dealing with incomplete data related to houses and limited amounts of data.

Total Rooms were created through feature engineering. Through model evaluation, it was observed that Random Forest gave the best performance for cross-validation amongst the three models considered. However, it performed poorly on the unseen test set.

There are, however, several limitations to the automated property valuation system. First, the model depends only on the information which was gathered in the dataset, and thus does not consider those features of the properties which have not been included into the dataset. Furthermore, the dataset includes only three suburbs, and thus the model cannot be immediately considered a representation of the whole Sydney housing market. This introduces a certain bias towards the properties included in the dataset.

Thus, the developed application should be considered a decision-support tool, and not an exact valuation system. Any predictions made by the system should be considered along with other information about the property.

However, had there been more time and resources, it would be possible to improve the project by collecting a significantly larger and more diverse dataset and testing various modelling methods.

Sydney Housing Price Prediction

Enter the property details to estimate its sale price.

Suburb

Blacktown

Bedrooms

3

Bathrooms

2

Parking

1

Land Size (m²)

500.00

Property Type

House

Predict Sale Price

Estimated Sale Price: \$1,035,298

OneDrive link:

[8.1D.ipynb](#)